

Machine Learning Models for Human *In Vivo* Pharmacokinetic Parameters with In-House ValidationFilip Miljković,^{*,▽} Anton Martinsson,[▽] Olga Obrezanova, Beth Williamson, Martin Johnson, Andy Sykes, Andreas Bender, and Nigel GreeneCite This: *Mol. Pharmaceutics* 2021, 18, 4520–4530

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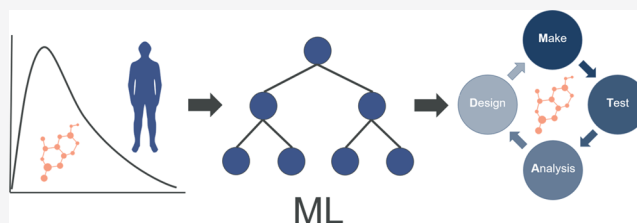
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ABSTRACT: Prior to clinical development, a comprehensive pharmacokinetic characterization of a novel drug is required to understand its exposure at the site of action and elimination. Accordingly, *in vitro* assays and animal pharmacokinetic studies are regularly employed to predict drug exposure in humans, which is often costly and time-consuming. For this reason, the prediction of human pharmacokinetics at the point of design would be of high value for drug discovery. Therefore, we have established a comprehensive data curation protocol that enables machine learning evaluation of 12 human *in vivo* pharmacokinetic parameters using only chemical structure information and available doses for 1001 unique compounds. These machine learning models were thoroughly investigated and validated using both an independent hold-out test set and AstraZeneca clinical data. In addition, the availability of preclinical predictions for a subset of internal clinical candidates allowed us to compare our *in silico* approach with state-of-the-art pharmacokinetic predictions. Based on this evaluation, three fit-for-purpose models for AUC PO ($R^2_{\text{test}} = 0.63$; $\text{RMSE}_{\text{test}} = 0.76$), C_{max} PO ($R^2_{\text{test}} = 0.68$; $\text{RMSE}_{\text{test}} = 0.62$), and Vd_{ss} IV ($R^2_{\text{test}} = 0.47$; $\text{RMSE}_{\text{test}} = 0.50$) were identified. Based on the findings, our machine learning models have considerable potential for practical applications in drug discovery, such as influencing decision-making in drug discovery projects and progression of drug candidates toward the clinic.

KEYWORDS: human pharmacokinetics, clinical data, machine learning, structure–property relationships, pharmacokinetic modeling, compound design



■ INTRODUCTION

Progression of a novel drug candidate toward the clinic requires comprehensive characterization of the pharmacokinetic (PK) behavior of the compound to ensure sufficient drug exposure at the site of action. The assessment and prediction of human PK play an integral role in drug discovery. However, PK remains one of the main contributing factors of compound attrition in clinical development.^{1–6}

In drug discovery, a suite of *in vitro* assays and animal PK studies are employed to assess the absorption, distribution, metabolism, and elimination (ADME) of a compound.² Hepatic matrices such as liver microsomes (LMs) and hepatocytes are utilized to assess the hepatic metabolic stability of the compound and allow extrapolation to the observed *in vivo* clearance (CL). Accurate prediction of total CL *in vivo* in animals from animal hepatocytes/LM increases confidence in the estimation of human CL from human hepatic matrices.³ Additional routes of elimination are further characterized by animal bile duct cannulated studies to assess biliary CL and urine collection studies to assess renal CL.² *In vitro* permeability assays and oral bioavailability determined in animals provide guidance on the expected human oral

exposure.⁴ For example, oral drug delivery projects typically aim to achieve high fraction absorbed and oral bioavailability (>30%) to achieve sufficient oral exposure in the clinic as bioavailability <25% is generally associated with variable PK in the clinic.⁴

Application of the drug metabolism and pharmacokinetics (DMPK) strategy defined at AstraZeneca² and the modeling techniques including physiologically based PK (PBPK) models to predict human PK and dose⁵ have, in the recent past, increased the number of successful AstraZeneca clinical candidates. In the past decade, 83% of drug discovery projects that reached clinical studies reported no PK issues.^{5,6}

While accuracy in predicting human PK has improved over the last decade, the approaches utilized at AstraZeneca⁵ require significant experimental data across species, *in vitro* matrices,

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and experimental formats, all with the appropriate number of repeats to understand the experimental error margin, *e.g.*, *in vitro* lipophilicity, solubility, permeability, metabolic stability, and protein binding in all animal and human species, *in vivo* PK across animals, drug–drug interaction data for human enzymes, and transporters. However, in the context of the design–make–test–analyze (DMTA) cycle, *in silico* approaches that use chemical structure information would be of significant utility as this information is available at early stages of a project when a chemist proposes a structure for synthesis.

To date, several *in silico* modeling studies that predict human and animal PK data from chemical structure have been reported.^{7–16} Compared to bioactivity prediction models, these studies utilize much smaller (and less homogeneous) data sets as *in vivo* data generation is generally more expensive and complex. In one of the earliest examples of human PK predictions, Lowe et al. built prediction models for rat and human LM CL and plasma protein binding using different machine learning methods.⁷ The CL models were successful at capturing overall data trends, whereas the models for plasma protein binding were deemed less informative.⁷ A more recent study based on a publicly available set of 1352 drugs⁸ modeled four human PK parameters (*i.e.*, volume of distribution at steady state ($V_{d_{ss}}$), CL, half-life, and fraction unbound in plasma) using four different regression methods coupled with a feature selection algorithm.⁹ The CL model demonstrated improved performance than those reported by other groups, most likely due to high data bias toward lower CL values. The $V_{d_{ss}}$ model performed equally well, although a more recent study reported models of similar performance that were built using an even more confined descriptor set.¹⁰ Kosugi and Hosea showed that machine learning models for CL provided a good alternative to traditional approaches (*e.g.*, *in vitro*–*in vivo* extrapolation), with potential to be used earlier in the drug discovery pipeline.¹¹ In another study, they demonstrated that the inclusion of *in vitro* data as features benefited the prediction of the oral area under the plasma time–concentration curve (AUC PO) in rats.¹² In addition, they suggested that a similar strategy could be applied to predict AUC PO for other preclinical species, as well as human.¹² Fagerholm et al. used an integrated strategy combining machine learning models, structural alert sets, and PBPK models to predict oral bioavailability in human.¹³ The authors concluded that the predictive accuracy (Q^2) of 0.50 (logit-transformed %) was sufficient to declare their models practical for further use.¹³

Predictive capabilities of novel machine learning approaches such as multitask deep neural networks or graph convolutional algorithms have also been explored to estimate *in vivo* PK data. For example, a Bayer study focused on the prediction of oral bioavailability as well as PO and intravenous (IV) drug exposure in rats.¹⁴ Several descriptor sets accounting for experimentally determined *in vitro* and physicochemical end points, predictions of *in silico* ADME models, and chemical structures represented as fingerprints or simplified molecular input line entry system (SMILES) strings were employed.¹⁴ This study showed that drug exposure after IV administration could be predicted equally well using either experimental or predicted properties as inputs. However, exposure after PO administration could be predicted more accurately using *in vitro* inputs.¹⁴ Another study utilized Merck data to build graph convolutional networks, which were subsequently compared to

a random forest algorithm.¹⁵ Performance of graph convolutional networks was better for the majority of the 31 data sets used in the study (*e.g.*, physicochemical, ADME, and *in vivo* dog and rat PK end points); however, for the *in vivo* end points, the improvement was marginal.¹⁵ An integrated multitask and transfer learning approach explored a pretrained model on over 30 million bioactivity data points to predict four human PK parameters (PO bioavailability, plasma protein binding rate, $V_{d_{ss}}$, and elimination half-life).¹⁶ Again, multitask learning demonstrated high predictability that, once integrated with transfer learning, resulted in greater accuracy and generalizability.¹⁶

Clearly, predictive models for *in vivo* animal and human PK data using chemical structure have become available in recent years. Our survey of the literature concludes that, while some of the end points (*e.g.*, $V_{d_{ss}}$) are easier to be modeled across different studies, others (*e.g.*, CL and bioavailability), however, are more challenging to predict, which is to be expected given the numerous factors that contribute to a compound's disposition.

The majority of studies so far dealt with only a few human PK end points that originated from pre-established data protocols, requiring little or no data intervention. Furthermore, validation and benchmarking were mostly limited to single data sources, which questions the applicability and validity of these models in future applications. While a large data set of human PK parameters has previously been reported,⁸ further expansion to include a greater number of end points, dose information, and error estimates of *in vivo* experiments would be highly valuable for a deeper understanding of human pharmacokinetics.

In this work, we have established a comprehensive data curation protocol to obtain a wide range of human *in vivo* PK parameters covering 1001 unique compounds that corresponded to 4491 compound–dose combinations from Food and Drug Administration (FDA) and European Medicines Agency (EMA) documents. These data formed the basis for the development of machine learning models, which used only chemical structure information and dose to predict individual PK end points. Models were thoroughly investigated and validated, utilizing both an independent test set and AstraZeneca's internal clinical data. Moreover, model performance was compared to PK predictions completed before first-time-in-human to define their applicability for internal use.

■ MATERIALS AND METHODS

Data Extraction. Human PK data were extracted from Elsevier's PharmaPendium PK module using its application programming interface (API) service (accessed Aug 2020).¹⁷ PharmaPendium gathers, extracts, and makes all of the historical data from FDA and EMA drug-approval documents readily available for data exploration.¹⁷ Initial variables to access API via Python were “species” (set to “human”) and “route of administration” (set to “oral” or “intravenous”). No additional variable restrictions were applied. Ultimately, we obtained a raw data set composed of 1 096 966 data records where each record detailed a single, aggregated PK measurement coming from an anonymized study.

Data Curation. Compound annotations were standardized using the RDKit¹⁸ and ChEMBL Structure Pipeline¹⁹ packages in Python. Standardization involved the removal of solvent molecules and salts, neutralization of acids and bases, removal of stereoisomers, conversion of isotopes, and representation

Table 1. Summary of Data Curation Steps^a

curation step	brief description	number of remaining data points
removal of missing SMILES	rows with missing SMILES (<i>e.g.</i> , drug combinations) were removed.	945 444
removal of study groups	healthy and patient study groups with retained PK functionality were kept.	881 409
SMILES standardization	SMILES standardization was performed using RDKit and ChEMBL Structure Pipeline packages.	866 833
removal of inadequate dose units	doses that could not be converted to milligram units, missing dose values, dose ranges, or values with qualifier signs were removed.	654 852
removal of inadequate PK parameter values	parameters with missing values, ranges, or values with qualifier signs were removed.	619 013
retention of single-dose administration	most commonly observed dosing regimen was kept.	419 078
removal of radiolabeled compounds	radiolabeled compounds were associated with unconventional PK study design and thus removed.	394 186
removal of high-molecular-weight compounds	compounds with $M_w > 750$ Da were removed to focus the study on small molecule pharmacokinetics.	384 481
removal of metabolites and enantiomers	only compounds that corresponded to provided chemical structures were retained.	292 599
removal of inadequate age groups	only adult population was retained, and mixed age groups were required to consist only of adults.	288 259
removal of certain assay technologies	assay technologies that were deemed unconventional for PK design (<i>e.g.</i> , bioassays, radioimmunoassays, <i>etc.</i>) were removed.	284 818
removal of inadequate concomitants	concomitants with the potential to impact PK measurements were removed (<i>e.g.</i> , grapefruit juice, high calcium diet, <i>etc.</i>).	252 963
removal of tissue-specific values	system analysis was preferred over tissue-specific results.	248 983
removal of inadequate text-free comments	manual revision of >5000 unique comments focused on the removal of data points, which indicated exclusion criteria specified above (<i>e.g.</i> , formulations, administration, patient information, <i>etc.</i>).	198 805

^aFor each consecutively applied curation step, its brief description and number of data points remaining after its application are reported.

canonicalization of aromatic structures. Although chemical standardization risks to exclude structural features relevant for PK prediction (*e.g.*, stereoisomerism or salt forms), majority of existing molecular descriptors, including those used herein, do not consider these for calculation. Additional categorical and numerical variables that are commonly present in regulatory documents such as study group information, concomitants, and protocol comments were part of the data set. This allowed us to define a rigorous data curation workflow to obtain higher quality of information for machine learning. Patient groups that may demonstrate altered PK profiles when compared to healthy individuals (*e.g.*, renal or hepatic impairment, *etc.*) were removed. We focused our analysis only on the adult population with both healthy volunteers and different patient groups with retained PK capabilities. This included indications where healthy volunteers cannot be enrolled (*e.g.*, oncology). Data from different study groups expanded our compound set and encompassed PK measurements from different phases of clinical trials. Next, data points with missing dose values, dose ranges, or qualifier signs preceding the value (*e.g.*, <, >) were removed. Only doses that could be converted to milligram (mg) units were retained. Milligram was the most frequently observed dose unit and was therefore set as a reference. All units denoting body mass measurements were converted, and 70 kilograms was taken as an average physiological human body mass. Studies that dealt with a single-dose administration that ranged between 0.1 and 10 000 mg (median value of 58 mg) were kept. Missing PK measurements, ranges, or values with qualifiers were removed. Radiolabeled compounds were identified as nonrepresentative of the conventional PK study design and were thus removed. Compounds with high molecular weight ($M_w > 750$ Da) were discarded. Metabolites and enantiomers that did not correspond to the provided chemical structure representation were removed. Measure-

ments captured by certain analytical assay technologies such as bioassays and radioimmunoassays were excluded. Studies involving concomitants indicating the potential impact on PK measurements (*e.g.*, grapefruit juice, high calcium intake, high-fat diet, *etc.*) were filtered out. Tissue-specific measurements were excluded from any further analysis. In addition, over 5000 unique free text comments were inspected for exclusion criteria detailed above (*e.g.*, inadequate formulations, administrations, patient information, *etc.*). Free text comments were often complementary to the discussed criteria and were thus added as an additional cleaning step. A step-by-step curation process is shown in Table 1.

With the above criteria, we have established an extensive data curation protocol, which produced high confidence in the PK values, suitable for any subsequent data exploratory analysis including machine learning. Data records were removed if they lacked crucial information such as dose, end point value, or compound information. However, for the majority of measurements, complete records were not provided to enable a more comprehensive cleaning procedure. For example, a PK measurement could be missing information about a study group, concomitant, analytical assay technology, or a combination of all three. Those studies and records were retained for further analysis.

Pharmacokinetic Parameters. The following nine original PK parameters were selected for the analysis: area under the plasma time–concentration curve (AUC) ($\mu\text{g}\cdot\text{h}/\text{mL}$), bioavailability (F) (%), maximum plasma drug concentration (C_{max}) ($\mu\text{g}/\text{mL}$), renal clearance of the drug from plasma (CL_R) (mL/min), apparent total body clearance of the drug from plasma (CL) (mL/min), elimination half-life ($t_{1/2}$) (h), time to reach C_{max} (t_{max}) (h), apparent volume of distribution at steady state (Vd_{ss}) (L), and apparent volume of distribution after nonintravenous administration (Vd/F) (L).

From these nine parameters, four were designated PO measurements (F , $C_{\max}$, $t_{\max}$, and Vd/F), two were IV measurements (CL and Vd_{ss}), and the remaining three were both PO and IV (AUC , CL_R , and $t_{1/2}$). In total, we investigated 12 PK parameters, and to denote the administration route of each, the PO/IV abbreviation was assigned as a suffix (e.g., AUC_{IV} , $C_{\max, PO}$, etc.).

Data Transformation and Aggregation. All PK parameter values were transformed by applying a decadic logarithm conversion. Doses were rounded to two decimal points if below 10 mg and zero decimal points if above this threshold. No logarithmic operation was performed on doses. Compound–dose combinations were defined, and parameter values were aggregated per each combination. For example, if a compound was reported with five different doses, then five compound–dose combinations were defined with their own set of PK parameters. For each compound–dose combination, median values were calculated per PK parameter and standard deviations were recorded. Standard deviations served a twofold purpose: first, to exclude parameter instances with high variance, and second, to estimate experimental errors for each PK parameter to benchmark model performances (see the [Contextualizing Model Errors and Experimental Variability](#) section). In the former case, the standard deviation criterion was applied, where a linear correlation between the mean values and standard deviations, for instance, where at least two replicates (i.e., measurements per compound–dose combinations) existed, was drawn.²⁰ From the calculated linear equation, values were excluded if the standard deviation was greater than two times the expected standard deviation at the given mean parameter value. Once aggregated and their variance explored, parameter value distributions were assessed and points beyond three standard deviations were discarded.

Chemical Space Analysis. Chemical space of clinical drugs was assessed and compared to a randomly sampled subset of 200 000 small molecules ($M_w < 750$ Da) from the ChEMBL database (version 27).²¹ Extended-connectivity fingerprint with bond diameter four (ECFP4, 1024-bit)²² from the RDKit package¹⁸ was used as a molecular representation. T-distributed stochastic neighbor embedding (t-SNE) method was applied to visualize the structural relationship of the two data sets.²³ As a nonlinear dimension reduction technique, t-SNE generates low-dimensional representations while preserving the local neighborhood of data points found in the original, high-dimensional space. In this approach, pairwise compound similarities are converted to joint probabilities and Kullback–Leibler divergence²⁴ between the high-dimensional data and joint probabilities of low-dimensional embedding is minimized. Different perplexity values were investigated and found to have very little influence on the visualizations. A value of 40 was chosen for the final visualization. Tanimoto distance was used as a metric to calculate the distance between instances in feature space. For convenience, kernel density estimation was applied to represent transformed ChEMBL data points in the t-SNE plot.

Molecular Representations. Several representations of molecular structures were generated using compound SMILES and evaluated using machine learning models. We grouped calculated descriptors into two-dimensional (2D) descriptors (OESelma molecular property descriptors²⁵ joined with clogP²⁶) and ADME/PK predictions (combination of *in silico* predictions from ADME²⁷ and *in vivo* rat PK assays). In addition, administered dose (Dose) was used as a single

descriptor to investigate its information content compared to other, more complex representations. Finally, a concatenated version of all three descriptor designs, termed ComboFP, was assembled separately. Altogether, we selected four descriptor designs for model generation.

2D Descriptors. The OESelma descriptors comprise 84 common one-dimensional (1D) and 2D molecular descriptors accounting for various physicochemical properties, such as the ring size, hydrogen bond donors, hydrogen bond acceptors, and polarity. The descriptors were generated using AstraZeneca's in-house script.²⁵ Additionally, clogP from Biobyte was added to the 2D descriptor set.²⁶

ADME/PK Predictions. ADME assay predictions were derived from nine single-task machine learning models, which were built from AstraZeneca ADME and physicochemical property data.²⁷ These nine end points were LogD, solubility (dried DMSO), Caco-2 intrinsic permeability, Caco-2 efflux ratio, human LM intrinsic clearance, human hepatocyte intrinsic clearance, rat plasma protein binding, human plasma protein binding, and fraction unbound in rat hepatocytes. The properties describe compound lipophilicity, solubility, permeability, intrinsic metabolic clearance, plasma protein binding, and incubational binding—molecular properties generally related to a wide range of *in vivo* end points. Decadic logarithm values were used for solubility, Caco-2 intrinsic permeability, Caco-2 efflux ratio, human LM, and hepatocyte intrinsic clearances. Logit transformation was applied to rat and human plasma protein binding and fraction unbound in rat hepatocytes.

Additionally, rat PK predictions were obtained using a multitask deep learning model based on Chemprop.²⁸ Chemprop is a message-passing neural network that extracts molecular representations from SMILES using graph convolutions in an end-to-end learning fashion.²⁸ *In silico* ADME/physicochemical predictions were concatenated with graph convolutions to define a final feature set for modeling. In-house AstraZeneca rat *in vivo* data composed of PK measurements for approximately 3500 compounds from various projects were used for model training and validation. Five rat PK parameters were predicted: F (PO), CL (IV), $C_{\max}$ (PO), $t_{1/2}$ (IV), and Vd_{ss} (IV). Outputs were given in decadic logarithm units (CL , $C_{\max}$, and Vd_{ss}), logit units (F), or were kept in original form ($t_{1/2}$).

In summary, nine ADME and five rat PK predictions for a total of 14 features constituted the ADME/PK prediction feature set to predict human PK.

Data Splitting Strategy. A customized data splitting strategy was applied during both cross-validation and hold-out test set definition to ensure better model generalizability and robustness. First, all compound–dose combinations were made part of the same split (or fold) to minimize the data leakage potential of the models and consequently their unrealistic performance. Second, data stratification was applied to all PK end points simultaneously to ensure that the same compound–dose combinations for all data splits across all PK parameter models were selected. To illustrate this, the training set was cross-validated using a k number of folds where each fold had a consistent set of compounds (including all corresponding doses) that were characterized by a proportional representation of all PK parameters. It is also worth noting that our data splitting strategy did not consider stratification of dose and PK parameter ranges with respect to their training sets. In addition, provided that 1001 clinical

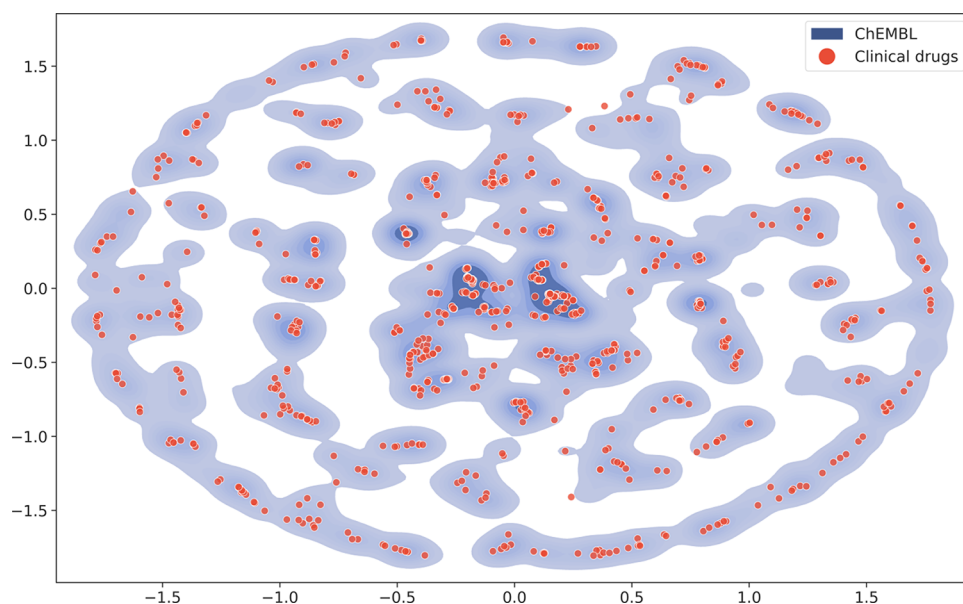


Figure 1. Chemical space of clinical drugs with curated PK data compared to known bioactive chemical space. The 2D t-SNE visualization shows the ECFP4 fingerprint space of clinical drugs (red points) and a random subset of ChEMBL compounds (kernel density estimate in blue). Color intensity of ChEMBL feature space scales with the density of clustered compounds.

drugs were chemically diverse (705 Bemis–Murcko scaffolds²⁹ with an average of 1.42 compounds per scaffold), no chemistry-based criteria were investigated for data splitting. Out of 132 scaffolds found in the hold-out test set, only 35 were present in the training set (or 26.7%), which corresponded to 55 compounds. These 35 scaffolds represented only 5.75% of the training set scaffolds, indicating a low overlap between the two sets.

With these requirements, we aimed to have representative cross-validation that could generalize prediction results for previously unseen chemistry and a wide range of doses and PK parameters. For all modeling approaches, 80% of the data were used for fivefold cross-validation and hyperparameter optimization. The remaining 20% were kept aside as a hold-out test set to evaluate the final models.

Machine Learning Methods. Regression models were generated using random forest (RF)³⁰ and eXtreme Gradient Boosting (XGBoost)³¹ algorithms. As reported earlier,¹⁵ deep learning did not provide an advantage over simpler machine learning methods in predicting *in vivo* PK data. It should be noted that only limited amounts of training data were available for modeling, which restricted the capacity of deep neural networks, making RF and XGBoost preferential methods for this study. For model building, training set samples were represented by a feature (see the [Molecular Representations](#) section). The numerical label was assigned for each PK task separately. Hyperparameter values for each method discussed below were exhaustively optimized via fivefold cross-validation.

Random Forest. RF is an ensemble of decision trees, each built from a bootstrapped sample of training data.³⁰ For the construction of individual trees, a random set of features was considered during node splitting.³⁰ Candidate values of 1, 5, and 10 were used to define the minimum number of samples at a leaf node (*min_samples_leaf*). Number of trees (*n_estimators*) was set to 20, 50, 100, or 500 and number of features considered when looking for the best split (*max_features*) was set to either all features or the square root of the total number

of features. RF method was implemented using the Scikit-learn package in Python.³²

XGBoost. XGBoost is an implementation of gradient boosting decision trees³¹ akin to the RF algorithm. In gradient boosting, new models are sequentially created to correct the errors made by prior models and then ensembled together to make the final prediction.³¹ Sequential models are created until no further improvements could be observed. Similarly to RF, values of 20, 50, 100, and 500 were explored for the number of trees (*n_estimators*). Learning rate (*learning_rate*) candidate values were 0.001, 0.01, and 0.1. Maximum depth (*max_depth*) values of 3, 5, 8, and 10 were used to denote the tree depth (*i.e.*, the length of the longest path from tree root to leaf). XGB regression method was applied using the XGBoost package in Python.³¹

Performance Metrics and Benchmarks. Both coefficient of determination (R^2) and root-mean-square error (RMSE) were used as performance metrics for model evaluation. The R^2 score presents the proportion of variance of label y explained by independent model variables. It provides an indication of goodness of fit and is defined as

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where y_i is the true value for the i -th sample, $\hat{y}_i$ is the predicted value for the i -th sample, $\bar{y}$ is the mean of the sum of all true values, and n is the number of samples. Scores typically have a range between 0 (poor performance) and 1 (perfect performance), although negative values of R^2 are obtained when a model mismatches the data trend. The RMSE calculates the square root of the quadratic mean of differences between true and predicted values. It is a measure of the spread of prediction errors around the line of best fit and is calculated by the following equation

$$\text{RMSE}(y, \hat{y}) = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}}$$

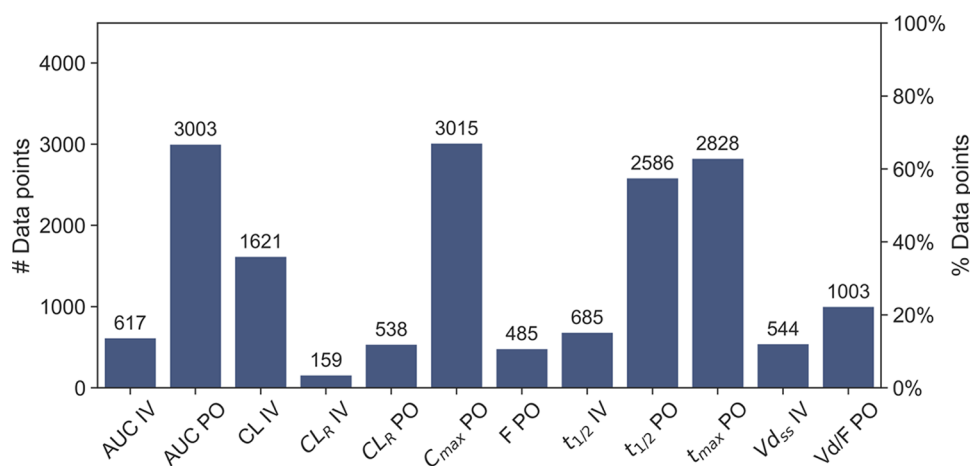


Figure 2. Distribution of pharmacokinetic parameters. Bar chart reports the number of measured compound–dose pairs per PK parameter. Absolute values are reported on top of each bar corresponding to the left-hand-side y-axis. Bars are scaled to the percentage of measured compound–dose combinations (right-hand-side y-axis). A height in 100% range corresponds to a PK parameter measured for all combinations.

The RMSE always takes a positive value where better model performance scales with lower scores and provides a point error estimate for individual predictions. In model generation, the RMSE score was used as an error metric during hyperparameter optimization.

Contextualizing Model Errors and Experimental Variability. Model performance was next put into the context of experimental errors, given that this determines an upper bound of model performance (which is of particular relevance for more variable *in vivo* experiments).³³ To this end, experimental errors were calculated for each PK parameter as two times the median value of standard deviation distribution.²⁰ In addition, the mean predictor was used as a simple baseline that always predicts a mean value of the training set for all test set instances. Mean predictions and experimental errors were regarded as points of lowest and highest practically achievable performances of our models, respectively (*i.e.*, from a completely random model reflecting the mean of the training set distribution to a near-perfect model only limited by an error of experimental procedure). Both experimental errors and mean predictor results were compared to model RMSE scores.

Model Validation on Internal Clinical Data. Performance of our models was additionally investigated using an internal AstraZeneca compendium of first-time-in-human measurements in the 2000–2020 period.⁵ The database is updated on a regular basis, which, at the point of this analysis, consisted of 126 drugs where the majority of discussed PK parameters were reported for each compound–dose combination. In addition, prediction values derived from DMPK modeling were provided alongside other data allowing for a side-by-side performance comparison with machine learning models.

RESULTS AND DISCUSSION

Clinical Drugs and Pharmacokinetic Data. Following the curation protocol, data for 12 PK parameters were extracted from PharmaPendium. A total of 1001 unique compound structures and 4491 compound–dose combinations were obtained, which formed the basis for further data analysis and modeling.

First, we investigated whether our set of clinical drugs was representative of a known chemical space of bioactive compounds. Figure 1 visualizes the joint chemical space of

1001 clinical drugs and a subset of randomly selected 200 000 small molecules from ChEMBL (~10% of the database, representing known bioactive chemical space) using t-SNE dimension reduction. As shown in the figure, [REDACTED]

We can conclude that the data set used in this study is diverse (and covers many modes of action and chemical classes), at least relative to the currently known bioactive chemical space, potentially resulting in broadly applicable predictive PK models.

We next aimed to understand the data composition with respect to end point annotations. Hence, Figure 2 summarizes the frequency of PK data points for all compound–dose pairs. The bar height represents the number of compound–dose combinations and is scaled to reflect the percentage of data completeness for each parameter. Values were available for a minimum of 3.5% (CL_R with 159 data points) and a maximum of 67.0% (C_{max} PO with 3015 data points) of all compound–dose pairs, indicating different levels of data incompleteness for PK parameters. A major underlying cause of data incompleteness is a practical one: majority of marketed drugs are intended for PO use and less commonly investigated intravenously and vice versa. This can be observed in Figure 2 where PO parameters are most commonly present. The four most frequent parameters (C_{max} PO, AUC PO, t_{max} PO, and t_{1/2} PO) are all PO variants and were measured in at least 57% of all compound–dose pairs, only to be followed by CL IV with 36% completeness. The difference was also observed between PO and IV variants of the same PK parameters where PO counterparts were most commonly measured. For example, AUC PO was reported five times more than AUC IV, CL_R PO three times more than CL_R IV, and t_{1/2} PO almost four times more than t_{1/2} IV.

With known PK parameters, one can apply equations to determine others (*e.g.*, AUC and dose can be used to calculate CL). However, this was not possible with the current data set for several reasons: first, the number of overlapping data points to evaluate the correlation of the calculated PK parameters was too few, and second, the end points that had a good overlap often correlated weakly (*e.g.*, Vd_{ss} IV and Vd/F PO). Parameter conversion additionally introduces the experimental error of parameter transformed, which in turn increases the

Model Performance. Models were built for all PK parameters using RF and XGBoost methods. Both methods were comparable for all tasks with a slight advantage of RF models. Thus, we further focus our discussion on RF performance for both cross-validation and hold-out test set based on the four different feature sets, as shown in Figures 3

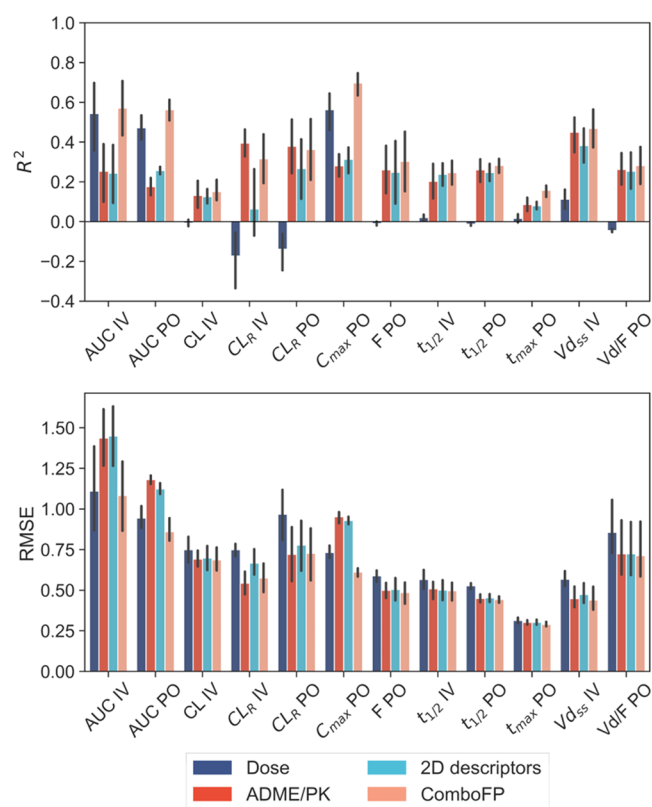


Figure 3. Fivefold cross-validation results across PK end points for different descriptor sets. Bar charts of R^2 (top) and RMSE (bottom) depict the average performance of fivefold cross-validation with vertical black lines on top of each bar showing uncertainty within a 95% confidence interval of the RMSE distribution. Different descriptor sets are color-coded following the legend below the charts.

and 4, respectively. In addition, we performed a prestudy to compare ECFP4 (1024-bits) fingerprints with ComboFP and observed no clear benefit of ECFP4, neither alone nor in combination. Moreover, ComboFP was able to outperform ECFP4 for several of the end points (e.g., Vd_{ss} IV, $\Delta R^2 = 0.18$; CL_R PO, $\Delta R^2 = 0.16$; C_{max} PO, $\Delta R^2 = 0.46$) and their combination often provided similar performance to ComboFP alone. For convenience of our analysis, we further investigate human PK data using only ComboFP and its components.

Figure 3 summarizes fivefold cross-validation results trained with a set of optimized hyperparameters intended for the final model building. Cross-validation results show good model robustness across all modeling tasks despite the challenging data splitting conditions. ADME/PK and 2D descriptors were often comparable and moderately predictive across PK

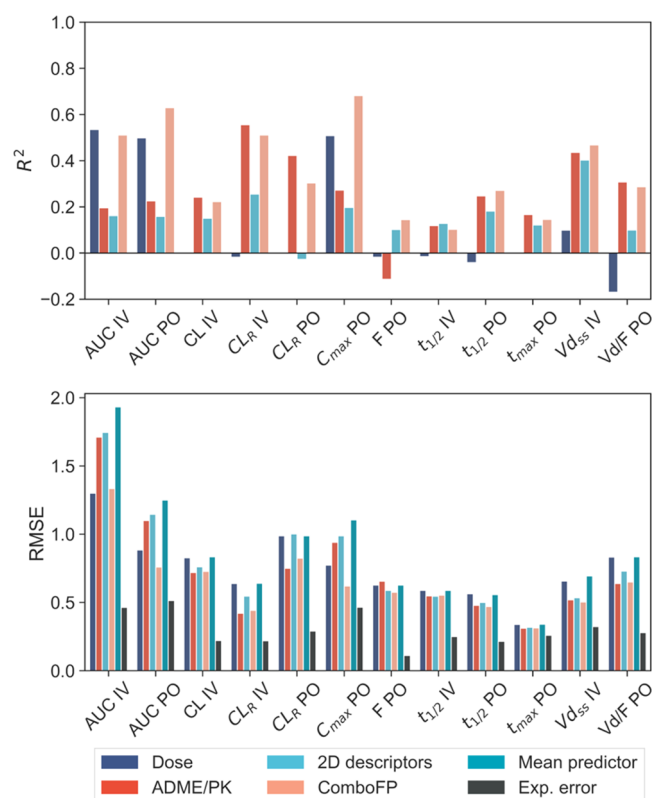


Figure 4. Hold-out test set results for different descriptor sets, compared to mean predictor and experimental variability. Bar charts of R^2 (top) and RMSE (bottom) depict model performance on a hold-out test set. Different descriptor sets, as well as two benchmark systems (mean predictor and experimental error), are color-coded following the legend below the charts.

parameters.

We next evaluated our model performance on a hold-out test set (20% of the curated data set not used for hyperparameter optimization and model building), as shown

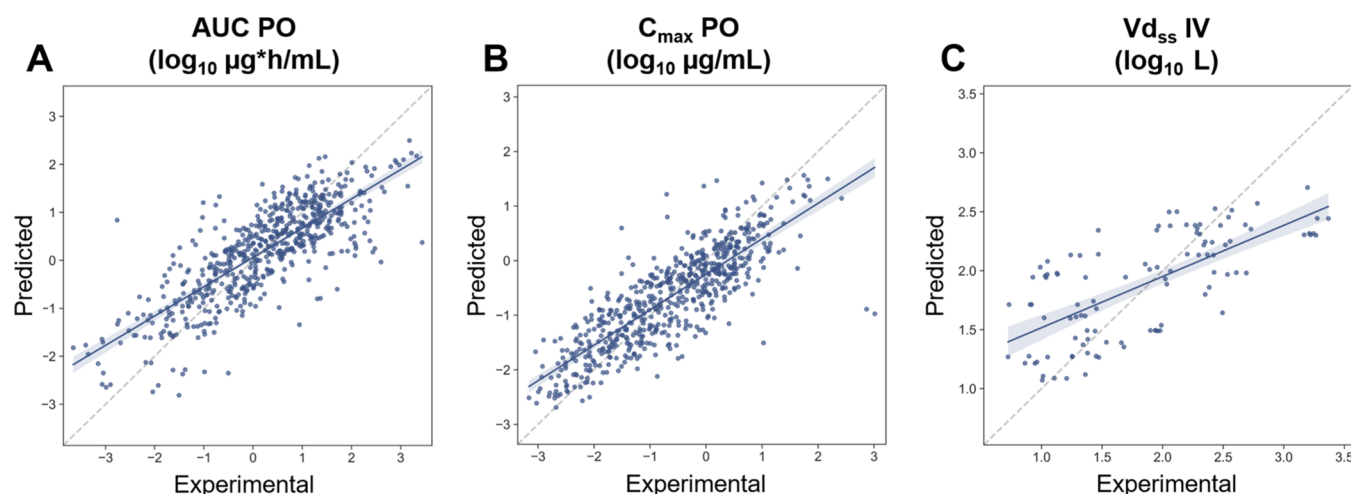


Figure 5. Correlation scatter plots of model performances. Scatter plots depicting the correlation between experimental (*x*-axis) and predicted (*y*-axis) values of a hold-out test set for AUC PO (A), $C_{\max}$ PO (B), and $V_{d_{ss}}$ IV (C) models are shown. For each scatter plot, the line of best fit (blue solid line with a 95% confidence interval band) and identity line (gray dashed line) are given.

Table 2. Model Performance on a Hold-Out Test Set^a

	N	R^2	RMSE	% <2-fold error	% <3-fold error	% <5-fold error
AUC PO	620	0.63	0.76	27.4	48.1	69.7
$C_{\max}$ PO	628	0.68	0.62	40.3	58.4	77.1
$V_{d_{ss}}$ IV	103	0.47	0.50	48.5	68.0	77.7

^aThe performance statistics for AUC PO, $C_{\max}$ PO, and $V_{d_{ss}}$ IV models on a hold-out test set are listed. For each model, number of tested compound–dose combinations, R^2 , RMSE, and percentage of combinations within two-, three-, and fivefold error thresholds are reported.

in Figure 4. The RMSE plot features additional values for mean predictor and experimental error as model benchmarks. A side-by-side comparison of cross-validation (Figure 3) and hold-out test set results (Figure 4) shows similar performance across all case scenarios, giving credence to the generalizability of our models. However, only five PK tasks achieved R^2 values above 0.40 (commonly accepted bottom threshold of good performance): AUC IV, AUC PO, CL_R IV, CL_R PO, $C_{\max}$ PO, and $V_{d_{ss}}$ IV. This performance was reached using ComboFP (AUC IV, AUC PO, and $C_{\max}$ PO), ADME/PK descriptors (CL_R PO), or either (CL_R IV and $V_{d_{ss}}$ IV). A different picture emerged, however, when considering model performances based on RMSE values. For example, the AUC IV model, which had a good R^2 value of 0.51 (ComboFP), resulted in an RMSE of 1.34, which is much closer to the performance of the mean predictor (1.94) than its experimental error (0.46), as displayed in Figure 4. Similar observations were made for the CL_R IV and CL_R PO models, where a narrow RMSE performance range was not much different from the corresponding mean predictor. Moreover, only 34 compound–dose combinations were part of the CL_R IV test set, which provided less confidence in the prediction results. This demonstrated the importance of calculating different performance metrics in model evaluation. On the other hand, AUC PO, $C_{\max}$ PO, and $V_{d_{ss}}$ IV retained RMSE performance that was much closer to the experimental error drawn from the data set (Figure 4). On the basis of R^2 and RMSE scores, the remaining models were insufficiently predictive. While some PK models showed good R^2 and RMSE performances, others were, despite promising R^2 values, not more predictive than the simple baseline setups.

Other data characteristics unfavorable to model building were also detected. For example, a $t_{\max}$ PO experimental error

of 0.26 was very close to the mean predictor performance of 0.34. This questions the usefulness of any $t_{\max}$ PO model derived from this data set, given the narrow distribution of data and relatively large error estimate. Another bias in the data stems from the fact that PK parameters of clinical drugs are often optimized within a preferred range of values during the development and this is reflected in the small number of unfavorable data points, which would otherwise be useful for machine learning. On the basis of the cross-validation (Figure 3) and hold-out test set (Figure 4) results, AUC PO, $C_{\max}$ PO, and $V_{d_{ss}}$ IV models built using ComboFP set were proceeded to further analysis.

Next, we visually analyzed the correlation of experimental versus predicted values for the hold-out set, given that numerical performance measures do not sufficiently reflect distributions of data across either of the axes. In this respect, Figure 5 shows experimental versus predicted values for AUC PO, $C_{\max}$ PO, and $V_{d_{ss}}$ IV models (subpanels A–C, respectively). It can be observed that the hold-out test set examples for these models had a wide range of experimental values, which were representative of the distribution for each parameter. Both AUC PO and $C_{\max}$ PO data spanned more than six logarithmic units and $V_{d_{ss}}$ IV almost four—ranges typical of all stages of drug development. Table 2 further corroborates the performance of the models. In terms of R^2 , $C_{\max}$ PO showed the highest value (0.68), followed by AUC PO (0.63), and finally $V_{d_{ss}}$ IV (0.47). For RMSE, $V_{d_{ss}}$ IV achieved the lowest value (0.50) of the three, followed by $C_{\max}$ PO (0.62) and AUC PO (0.76). Compared to AUC PO and $C_{\max}$ PO, the $V_{d_{ss}}$ IV data were one-sixth of the size, yet produced a good and robust model. Moreover, for all three models, the percentage of predictions within two-, three-, and fivefold errors (± 0.30 , ± 0.48 , and ± 0.70 logarithmic units,

Table 3. Model Performance on the Internal Clinical Data^a

	N	R^2	RMSE	% <2-fold error	% <3-fold error	% <5-fold error
AUC PO	58	0.32	0.71	32.8	50.0	63.7
$C_{\max}$ PO	60	0.53	0.65	30.0	50.0	73.3
$V_{d_{ss}}$ IV	5	0.80	0.18	100.0	100.0	100.0

^aPerformance statistics for AUC PO, $C_{\max}$ PO, and $V_{d_{ss}}$ IV models on the AstraZeneca clinical drug set are listed. For each model, number of tested compound–dose combinations, R^2 , RMSE, and percentage of combinations within two-, three-, and fivefold error thresholds are reported.

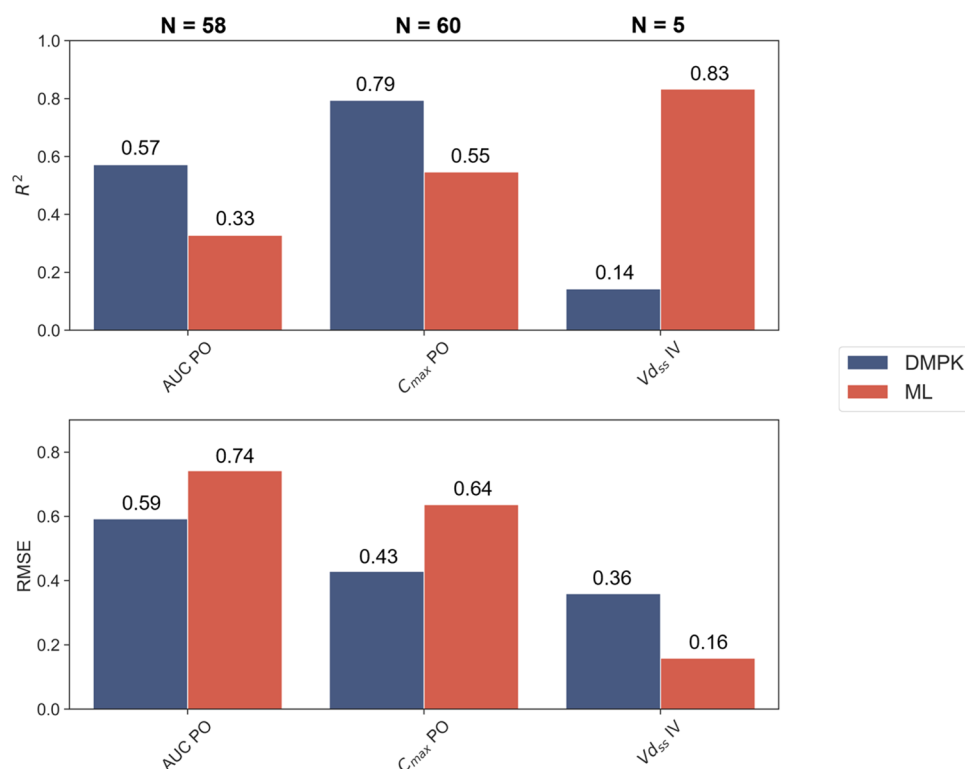


Figure 6. Performance comparison of DMPK and machine learning predictions. Bar charts of R^2 (top) and RMSE (bottom) results of DMPK (blue) and machine learning (ML) modeling (orange) for AUC PO, $C_{\max}$ PO, and $V_{d_{ss}}$ IV models on the basis of internal clinical data are shown. On top of each bar, a corresponding value of the performance measure is given. Above the plot area, the number of evaluated compound–dose combinations per PK parameter is displayed.

respectively) was calculated (Table 2). For example, the $V_{d_{ss}}$ IV model (as the weakest of the three in R^2 terms) had a prediction percentage within two-, three, and fivefold errors of 48.5, 68.0, and 77.7%, respectively. On the other hand, $C_{\max}$ as the best-performing model had equivalent values of 40.3, 58.4, and 77.1% for the corresponding error margins. This concludes that the correlation plots and additional performance measures further exemplified the predictiveness of our models, advancing all three to the next step of validation with AstraZeneca clinical candidates.

Model Performance on Internal Clinical Candidates.

Performance results shown above demonstrated that the three best-performing models could successfully be applied to a range of chemically diverse compounds, as exemplified by historic clinical drugs. However, to ensure fitness for their proper use within AstraZeneca, a validation on our internal set of clinical candidates was also required. Therefore, we accessed an internal compendium of clinical candidates (which to this date counts 126 drugs) with associated experimental and state-of-the-art predicted PK data, none of which was present in the training data sets. A confined subset of PO and IV drugs with a molecular weight below 750 Da had values reported for the discussed PK parameters. In comparison to 26.7% of hold-out

test set scaffolds, which overlapped with the training data, internal clinical candidates had only six scaffolds (or 5.2%) present in the training set, indicating a more challenging prediction task for the models. Table 3 discusses the number of available data points and performance metrics for each model. Validation results showed similar model performance despite the reduced compound set. Here, AUC PO predictions had a considerably lower R^2 score but a similar RMSE value compared to the hold-out test set results. Similar results were obtained by $C_{\max}$ PO predictions where R^2 performance dropped by 0.15 units and RMSE remained relatively unchanged. Both AUC PO and $C_{\max}$ PO had a third and a half of the predicted values within two- and threefold error margins, respectively. Performance of the $V_{d_{ss}}$ IV model was outstanding with $R^2 = 0.80$, RMSE = 0.18, and all predicted values fully within the three error margins. However, given that only five compound examples were available for the $V_{d_{ss}}$ IV end point no firm conclusions could be made on the basis of internal clinical data. Therefore, we can conclude that, although the hold-out test set and internal clinical compounds have different data set sizes and chemical composition, their prediction results are relatively consistent (in terms of different data sizes, RMSE is a preferred metric).

Comparison to Preclinical PK Predictions. Additionally, the internal database featured PK predictions (for only a subset of clinical candidates with reported experimental values), which were derived within DMPK by extrapolating PK parameter values of animal species to human or from *in vitro* assays.^{2,5} Predictions of our three machine learning models (AUC PO, $C_{\max}$ PO, and $V_{d_{ss}}$ IV) were compared to those predicted internally (Figure 6). In terms of R^2 results, the internal DMPK predictions outperformed machine learning for AUC PO (45 compounds) and $C_{\max}$ PO (59 compounds), where the difference was ~ 0.24 for both cases. However, RMSE values revealed very little difference between the two, with the DMPK predictions demonstrating lower scores for both AUC PO (Δ AUC PO = 0.15) and $C_{\max}$ PO (Δ $C_{\max}$ PO = 0.21). On the other hand, the $V_{d_{ss}}$ IV machine learning model was superior to that determined within DMPK, showing R^2 and RMSE improvements by 0.69 and 0.20, respectively, although only for five data points. Altogether, internally validated machine learning models appear fit-for-purpose at the point of design within AstraZeneca. Together with the standard DMPK prediction approaches,^{2,5} these models may provide an early complementary assessment of human PK parameters and influence decision-making when progressing drug candidates toward the clinic.

CONCLUSIONS

In this proof-of-concept study, we have—to the best of our knowledge—established the first comprehensive protocol for the curation of human PK data, which enables the machine learning-based prediction of compound PK parameters using only chemical structure information and dose. An expert-driven curation protocol ensured that only high confidence data for human PK parameters were utilized for model generation. In a chemical space analysis, we have shown that the structural space covered by curated clinical drugs corresponded to that of the currently known bioactive small molecules, indicating a broad applicability domain of successfully derived machine learning models. To validate this in practice, different descriptor designs were investigated using customized data splits that achieved high model robustness for all case scenarios. However, only three fit-for-purpose models for AUC PO, $C_{\max}$ PO, and $V_{d_{ss}}$ IV that were derived from the descriptor set consisting of a dose, 2D descriptors, and ADME/PK predictions (ComboFP) were identified based on the cross-validation. These models were further successfully validated using a hold-out test set and AstraZeneca clinical data. Despite their different sample sizes and chemical composition, the model performance was comparable for both the hold-out test set and internal clinical candidates. For the remaining nine PK parameters, no fit-for-purpose models could be derived on the basis of investigated molecular representations. Inherent data characteristics, such as data incompleteness, noise, and absence of “negative” examples, may have contributed to this finding.

Although the state-of-the-art DMPK predictions performed comparably or better for a subset of internal clinical compounds, their development and validation are much more time-demanding. Future use of machine learning-assisted PK models could be imagined alongside current approaches, where their complementary use could improve the success and rate of candidate drugs reaching clinical development. For example, our machine learning models implemented internally can be used at the point of design (e.g., within the DMTA

cycle) to prioritize compounds with desirable PK predictions whose findings could later be consolidated by more comprehensive modeling.

As we continue to grow our understanding of the human PK data, we aim to further expand our chemoinformatics toolbox (e.g., to include other molecular representations, machine learning approaches, etc.) and eventually deliver suitable predictive models for other PK parameters. Our current models show great potential to guide *de novo* generative models to design compounds with improved PK properties, aid individual project decisions, improve predictions of safety-related end points (e.g., $C_{\max}$ in predicting physiological effects of off-target activity, or AUC in predicting cytotoxicity, necrosis, or other cellular damage end points), as well as to assist candidate prioritization toward the clinic.

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Notes

The authors declare no competing financial interest.

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